

Assignment #3: Iterative Refinement / FP Theorem

Due: November 17, 2024 at 11:45 p.m.

This assignment is worth 10% of your final grade.

Warning: Your electronic submission on *MarkUs* affirms that this assignment is your own work and no one else's, and is in accordance with the University of Toronto Code of Behaviour on Academic Matters, the Code of Student Conduct, and the guidelines for avoiding plagiarism in CSCC37.

This assignment is due by 11:45 p.m. November 17. If you haven't finished by then, you may hand in your assignment late with a penalty as specified in the course information sheet.

- [20] 1. Consider the Iterative Refinement (IR) algorithm discussed in lecture and tutorial:

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Compute  $\hat{x}^{(0)}$  by initially solving  $Ax = b$ .  
For  $k = 0, 1, 2, \dots$  until “convergence” or “convergence failure”  
    compute  $r^{(k)} = b - A\hat{x}^{(k)}$   
    solve  $Az^{(k)} = r^{(k)}$  for  $\hat{z}^{(k)}$   
    update  $\hat{x}^{(k+1)} = \hat{x}^{(k)} + \hat{z}^{(k)}$   
end for
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- (a) Give details of two tests for “convergence”, and two tests for “convergence failure”. One of your tests for convergence failure must efficiently assess the condition of A , using the matrix condition number estimation bound discussed in tutorial.
- (b) We gave an intuitive explanation of why this algorithm could improve the initial approximate solution $\hat{x}^{(0)}$, but we were vague on the conditions required for convergence. In this question you will derive precise requirements for IR convergence.
- Consider $z^{(k)}$ and $\hat{z}^{(k)}$, and the residual systems:

$$Az^{(k)} = r^{(k)} \quad \text{and} \quad (A + E)\hat{z}^{(k)} = r^{(k)}, \quad (1)$$

where E is a perturbation matrix representing the round off error accumulated when the residual system is solved using floating point arithmetic. Starting with (1), derive a recurrence showing how the absolute error in the $(k+1)$ -st iterate $\|\hat{x}^{(k+1)} - x\|$ is bound by an amplification of the absolute error in the k -th iterate $\|\hat{x}^{(k)} - x\|$. Argue that the magnitude of this amplification is dependent on the condition of A , and is fractional – hence we will have convergence – if A is well-conditioned.

- [5] 2. When the IR algorithm in #1 is converging, the size of the residual should eventually begin to decrease as the iteration proceeds; i.e.,

$$\|r^{(k)}\| = \|b - A\hat{x}^{(k)}\| \rightarrow 0 \quad \text{as } k \rightarrow \infty$$

(This, in fact, should be one of your convergence tests in #1(a).) The i -th component of the residual $r_i^{(k)}$ is calculated as:

$$r_i^{(k)} = b_i - \sum_{j=1}^n a_{ij} \hat{x}_j^{(k)} \quad (2)$$

As the iteration is converging, there is potential for subtractive cancellation in (2) when

$$|b_i| \approx \left| \sum_{j=1}^n a_{ij} \hat{x}_j^{(k)} \right|$$

and the signs of these two terms are either both positive or both negative. Is there an algebraic equivalent to (2) which is less sensitive to subtractive cancellation? If not, how can we protect from subtractive cancellation when evaluating (2)?

- [10] 3. Rewrite the IR algorithm given in #1 to show how $PA = LU$ factorization, forward elimination, and back solve are used when solving systems involving A (i.e., both in the initial computation of $\hat{x}^{(0)}$, and in subsequent computations of $\hat{z}^{(k)}$ within the loop). Do this in such a way that the IR algorithm is *optimally efficient* with regard to operation counts.

After rewriting the algorithm, give a formal analysis of its flop count (floating point operation count), using big-oh notation in a similar fashion as we did in lecture when analyzing the complexity of Gaussian Elimination. You may assume that the algorithm converges in M iterations. Your complexity equations will depend on both M and n . (Note that M is relevant to one of your convergence failure tests in #1(a).)

- [15] 4. Consider the functions $f(x) = 1 - 1/(2x)$ and $g(x) = 2x(1 - x)$.
- (a) How many roots does f have? Are the roots of f fixed-points of g ? Are there more fixed points of g than roots of f ? **Justify your answers.**
 - (b) For f and g as defined above, identify $\tilde{x} = g(\tilde{x})$ as a second-form fixed point problem corresponding to the root finding problem $f(\tilde{x}) = 0$. What is the auxiliary function $h(x)$ in this second form?
 - (c) Using the *Fixed Point Theorem* (FPT) – and *only* the FPT – determine the largest possible interval for which the the fixed-point iteration $x_{k+1} = 2x_k(1 - x_k)$ is *guaranteed* to converge. Is the interval you identified above the *maximum* guaranteed interval of convergence? In other words is it not possible, with additional analysis combined with the FPT, to find an even larger guaranteed interval of convergence? (See also #5 below.)

- [10] 5. Consider the iteration $x_{k+1} = e^{-x_k}$ for computing the fixed points of $g(x) = e^{-x}$, the first-form fixed point problem equivalent to the root finding problem $f(x) = x - e^{-x} = 0$. Use the FPT – combined with additional analysis – to determine the *maximum* guaranteed interval of convergence for this iteration.

Caveat: In lecture we used the FPT to analyze this iteration, but we did **not** determine the *maximum* guaranteed interval of convergence.

[total: 60 marks]